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***Strategic Launch Analysis for New York
City Robotaxi Deployment***



Prepared By:

Prateeksha Mehta

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1. Executive Summary

The objective of this project is to develop a data-driven launch strategy for a Robotaxi service in New York City. We specifically target transit deserts, which are areas where residents lack adequate access to existing public transportation. By utilizing advanced analytics and machine learning, we identified specific geographic zones where autonomous mobility can provide the highest commercial value while addressing critical gaps in social equity.

Our methodology integrated three diverse datasets, including demographic data from the United States Census Bureau, public infrastructure data from the MTA, and over 10 million real-world ride-hail records from the Taxi and Limousine Commission. We employed a dual-model approach, using a Random Forest Classifier to identify high-priority launch zones and an XGBoost Regressor to forecast monthly trip demand and potential revenue for these areas.

Key findings from our analysis reveal that the primary drivers of Robotaxi demand in New York City are not household income but rather a lack of personal vehicle ownership and the presence of populations with physical disabilities. Our primary strategic model achieved an 89 percent accuracy rate in classifying high-priority zones, allowing us to generate a definitive top 10 list of pilot launch locations.

Based on our findings, we recommend a phased deployment strategy starting with these 10 high-potential neighborhoods. At a conservative 5 percent market capture rate, these locations represent a significant multi-million-dollar annual

revenue opportunity. This project demonstrates that by bridging the gap between transportation infrastructure and community needs, a Robotaxi service can achieve both financial profitability and long-term social impact.

2. Introduction

2.1. Project Overview and Business Case

New York City remains one of the most complex urban environments in the world regarding transportation. While the city boasts an extensive subway and bus network, significant portions of the outer boroughs and even certain areas within Manhattan are classified as transit deserts. These are regions where residents live more than a half mile away from the nearest subway station. For individuals without access to a private vehicle, particularly those with physical disabilities or those working late shift service jobs, this lack of infrastructure creates a significant barrier to economic and social mobility.

This project introduces a strategic solution to this problem through the deployment of an autonomous Robotaxi service. Unlike traditional ride-hailing platforms that often cluster in high-income Manhattan neighborhoods, our business case focuses on a targeted launch in these underserved transit deserts. By utilizing autonomous vehicle technology, we can lower operational costs and provide a reliable, door-to-door transit alternative that operates 24 hours a day. This initiative aims to prove that a modern transportation company can achieve significant financial returns while simultaneously fulfilling a critical social need.

2.2. Research Questions and Objectives

To ensure a successful and data-driven launch, this study seeks to answer three primary research questions that guide our analytical process. First, we ask where the most critical gaps in New York City transit infrastructure exist based on physical distance to existing MTA stations. Second, we investigate which demographic and economic factors most accurately predict the demand for ride-hailing services in these identified gaps. Finally, we ask if machine learning models can be trained to successfully prioritize specific census tracts for a pilot launch program.

The objectives of this report are to provide a clear roadmap for decision makers. These objectives include the identification of the top ten launch sites, a detailed forecast of potential monthly trip demand using advanced regression techniques, and a financial impact analysis that demonstrates the projected return on investment. By answering these questions, we move from a theoretical business idea to an actionable deployment strategy backed by millions of data points.

3. Literature Review

3.1. Current State of Autonomous Vehicle Economics

Autonomous vehicles are becoming an important topic in the ride-hailing industry because they could change how companies manage cost, pricing, and fleet operations. In traditional ride-hailing, driver compensation is one of the highest operating costs. Robotaxis may reduce this cost over time, but the economic

benefit is not automatic. A successful robotaxi service still needs to manage vehicle supply, charging, maintenance, dispatching, customer demand, and local regulations.

Lian and van Ryzin (2024) explain that autonomous vehicles can create value in ride-hailing when fleet size, dispatching, and pricing are designed around market demand. Freund, Lobel, and Zhao (2025) also emphasize that autonomous vehicle platforms must carefully decide how many vehicles to deploy and where to position them. These studies support our project's approach of using data to identify specific New York City neighborhoods for deployment instead of recommending a broad citywide launch.

Recent market developments show that robotaxi services are moving from testing to early commercial operations. For example, the California Public Utilities Commission approved Cruise and Waymo to charge fares for passenger service in San Francisco in 2023. Although San Francisco is different from New York City, this example shows that robotaxis are becoming a real regulated transportation service.

Cost and pricing are also key factors. Research from the International Energy Agency and the U.S. Department of Energy helps estimate electric vehicle ownership and operating costs, while the National Renewable Energy Laboratory highlights the added planning needs for automated electric mobility systems. Obi's pricing research on Waymo, Uber, Lyft, and Tesla-related competition also suggests that autonomous ride pricing can vary by market and demand level.

These sources support our use of conservative cost assumptions and demand-based pricing in the ROI model.

Overall, the literature shows that robotaxi economics depends on more than the technology itself. Profitability depends on where vehicles are placed, how demand is predicted, how prices are set, and how operating costs are managed. This supports our project's data-driven strategy for selecting high-priority NYC launch zones.

3.2. Transit Deserts and Social Equity in Urban Mobility

Although New York City has a large public transportation system, transit access is not equal across all neighborhoods. Some residents live close to subway stations, while others must travel much farther to reach reliable transit. These areas are often described as transit deserts. In our project, we focus on neighborhoods where residents are more than a half mile from the nearest subway entrance because this distance can create major barriers for people without access to a private vehicle.

The Urban Institute explains that autonomous vehicles could either improve or worsen transportation equity depending on how they are deployed. If robotaxis are placed only in wealthy or tourist-heavy areas, they may mainly serve people who already have many transportation options. However, if they are deployed in underserved neighborhoods, they could improve access to jobs, healthcare, education, and daily activities.

Fatima (2024) also discusses equity concerns related to AV adoption, especially for low-income groups, disabled riders, minority communities, and people facing transportation barriers. Wu et al. (2021) similarly show that vehicle automation may benefit transport-disadvantaged populations, including older adults and people with disabilities. These studies support our use of variables such as disability percentage, vehicle ownership, public transit use, poverty rate, and commute time.

Ride-hailing research is also relevant to this project. Li et al. (2022) suggest that ride-hailing can help fill mobility gaps in areas where public transit is limited or inconvenient. This supports our use of TLC ride-hailing data as a proxy for potential robotaxi demand. If certain neighborhoods already rely heavily on ride-hailing, they may also be strong candidates for robotaxi service.

Together, these studies support our focus on NYC transit deserts rather than only high-income or high-tourism areas. The literature shows that robotaxi demand may be connected to real mobility needs, such as a lack of vehicle ownership, distance from subway access, and disability status. This makes robotaxi deployment both a business opportunity and a potential tool for improving transportation access.

3.3. Technical Frameworks for Demand Prediction

A robotaxi launch strategy needs a technical framework that can estimate where demand is likely to be strongest. For this project, we combine census

demographics, public transit infrastructure, geospatial distance measures, and historical ride-hailing data. This allows us to evaluate New York City at the neighborhood level instead of making general assumptions about the whole city.

The studies by Lian and van Ryzin (2024) and Freund et al. (2025) support this type of modeling because autonomous ride-hailing platforms depend on matching vehicle supply with passenger demand. This is why our project first identifies transit deserts and high-need areas, then predicts potential trip demand and ROI.

Our method starts by calculating each census tract's distance to the nearest subway entrance. We then combine this with demographic variables such as vehicle ownership, disability status, income, public transit use, service occupation percentage, and commute time. These variables help us measure both demand potential and mobility needs.

We use a Random Forest Classifier to identify high-priority launch zones because it can handle nonlinear relationships between different factors. For example, a strong robotaxi candidate may have low vehicle ownership, long distances to subway access, high disability rates, and existing ride-hailing demand. We then use XGBoost to predict trip volume, which becomes the basis for our ROI model.

The equity and ride-hailing literature also supports our feature selection. Fatima (2024), Wu et al. (2021), and Li et al. (2022) show that transportation demand is shaped by access, disability, vehicle ownership, and local transit gaps. Cost and

pricing references from the IEA, DOE, NREL, and Obi help connect predicted demand to revenue, operating cost, and financial feasibility.

Overall, the literature supports our project's workflow: identify transit deserts, measure mobility needs, classify high-priority launch zones, predict trip demand, and evaluate ROI. This gives our team a practical, data-driven framework for recommending where robotaxis should launch first in New York City.

4. Data Acquisition and Preparation

4.1. Overview of Data Sources

The foundation of this project relies on the integration of several diverse datasets that capture the demographic, infrastructure, and market landscape of New York City. The first category includes United States Census Bureau records from the five-year American Community Survey. These files provide granular information on census tracts, covering key variables such as vehicle ownership, disability status, and employment types. To properly map this information, we utilized the `nyct2020` dataset, which contains the official geographic boundaries for every census tract within the city.

The second category focuses on transportation infrastructure and city planning. We utilized official datasets from the city's open data portal to identify the exact coordinates of every bus shelter. Additionally, we incorporated Metropolitan Transportation Authority data to locate subway entrances. By combining these

infrastructure layers, we were able to see exactly where public transit is available and where the gaps remain.

The third category consists of market and movement data from the Taxi and Limousine Commission. We utilized the taxi zones shapefiles and the taxi lookup tables to ensure that our analysis aligned with official city transit sectors. This data allowed us to connect our neighborhood demographics with millions of real-world trip records. By merging these various files into a single master dataset, we created a comprehensive view of how people move through New York City and where a Robotaxi service is most needed.

4.2. Data Cleaning and Feature Engineering

Raw data often contains noise and missing values that can negatively impact the performance of machine learning models. We developed a custom Python function to clean and standardize the census files, ensuring that complex technical identifiers are converted into readable neighborhood characteristics. During this process, we identified and handled missing data points by using median imputation, which preserves the overall statistical distribution of the New York City population.

To make the data more useful for our models, we performed feature engineering by calculating new metrics. For example, we converted raw population counts into percentages to allow for fair comparisons between small and large neighborhoods. We also combined various disability categories into a single total

disability metric to better understand the overall need for accessible transportation in each zone.

4.3. Geospatial Integration and Mapping

The most critical step in our data preparation was the geospatial join. We used geographic information system tools within Python to link the demographic data of each census tract to its physical location in the city. By calculating the distance from the center of each neighborhood to the nearest subway entrance, we were able to mathematically define which areas are true transit deserts.

We also aligned these census tracts with official taxi zones to ensure our demand forecasts matched real-world trip data. This coordination of different map layers allowed us to create a master dataset where every row represents a specific neighborhood equipped with its demographic profile, its distance from public transit, and its historical demand for rides. This master file serves as the primary input for the classification and regression models used later in the study.

5. Exploratory Data Analysis (EDA)

5.1. Correlation Analysis and Identifying Drivers of Mobility

After mapping the physical gaps, we conducted a correlation analysis to understand the demographic profile of people living in these areas. We used a correlation matrix to see which neighborhood characteristics move in the same direction as ride-hailing demand. Our analysis showed a very strong positive

correlation between a lack of personal vehicle ownership and the frequency of ride-hailing trips. This suggests that for many New Yorkers, services like Uber or a future Robotaxi are not luxuries but essential tools for daily life.

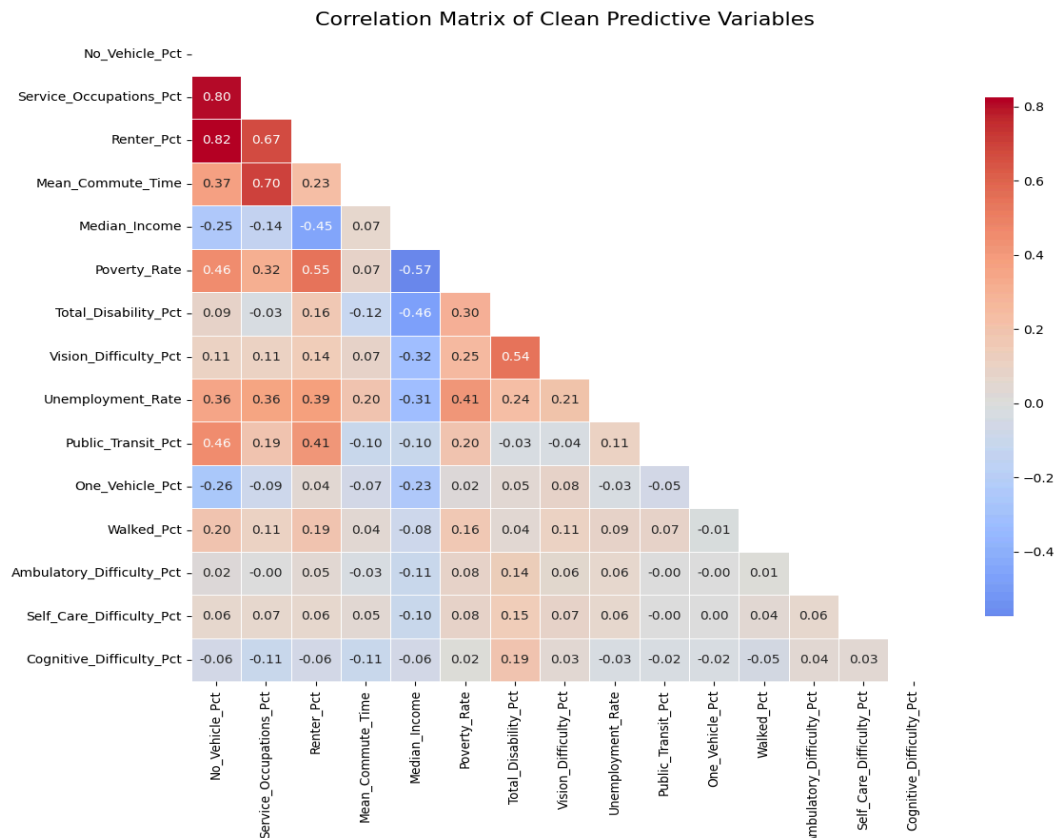


Figure 1. Correlation Matrix of Clean Predictive Neighborhood Variables

The correlation matrix provides a statistical foundation for our predictive modeling by highlighting the strong relationships between specific neighborhood characteristics. Our analysis indicates that a lack of personal vehicle ownership and a high percentage of residents with disabilities have the strongest positive correlations with ride demand. Conversely, the matrix shows that median

household income has a much weaker relationship with transit needs in these specific areas than traditionally assumed. These findings allow us to prioritize the most impactful variables in our machine learning models, ensuring that our launch strategy is built on the most relevant data drivers.

We also discovered a significant correlation between high disability rates and transit desert locations. This is a critical finding because it highlights a major gap in the current MTA system regarding accessibility. Furthermore, we observed that median household income was not as strong a predictor for ride demand as we originally expected. This indicates that the need for reliable transportation transcends income levels in areas where the subway is out of reach. These insights allowed us to move away from basic demographic assumptions and toward a more sophisticated model based on actual human mobility needs.

5.2. Analyzing the Scale and Drivers of Transit Deserts

After identifying the demographic drivers of mobility, our analysis shifted toward quantifying the physical infrastructure gaps across New York City. This phase of the Exploratory Data Analysis (EDA) focused on measuring the Transit Desert effect through two distinct lenses: the city-wide distribution of need and the direct impact of subway proximity.

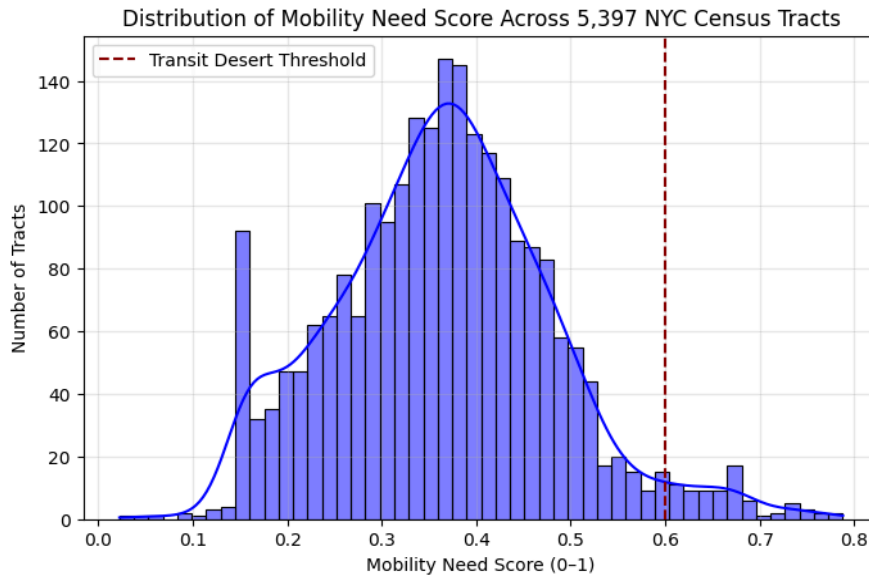


Figure 2. Distribution of Mobility Need Scores Across NYC Census Tracts

The distribution of mobility need scores across all five thousand census tracts reveals a significant concentration of neighborhoods with high transit requirements. As shown in this histogram, a large number of tracts fall into the upper percentiles of the scoring range, indicating that the need for alternative transit is not a localized issue but a city-wide challenge. This visual evidence justifies the commercial viability of a Robotaxi service by proving that there is a consistent and high-volume market for autonomous mobility across many New York City neighborhoods.

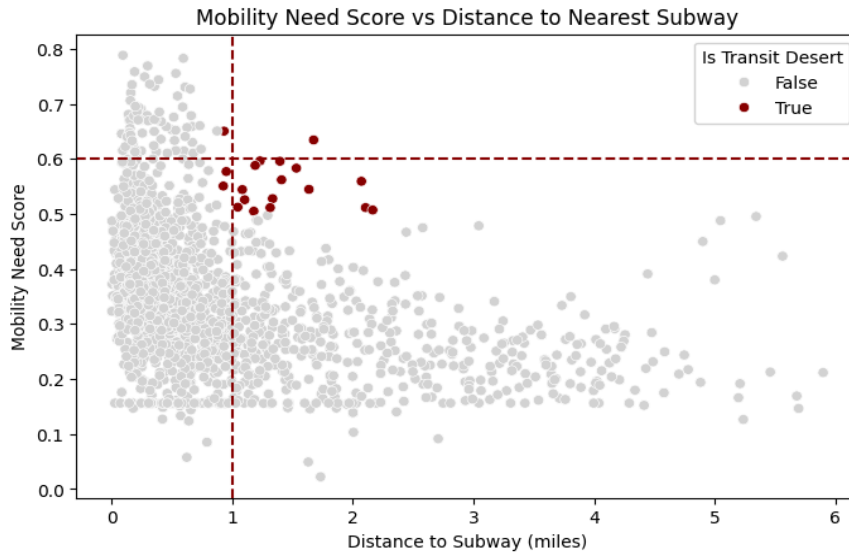


Figure 3. Relationship Between Subway Proximity and Mobility Need

This scatterplot establishes a direct mathematical link between the physical distance to public infrastructure and the resulting mobility needs of the population. As the distance to the nearest subway entrance increases beyond the half-mile mark, we observe a clear upward trend in the mobility need score. This visualization confirms that geographic isolation from the rail network is a primary driver of transit desert conditions. By identifying the tracts that sit in the upper right quadrant of this chart, we can precisely target the areas where a Robotaxi service will provide the most immediate relief to residents.

We also looked at the relationship between bus shelter locations and these subway gaps. Surprisingly, many areas with poor subway access also have a low density of bus shelters, which compounds the transportation problem for residents. These findings confirmed our initial hypothesis that there is a massive and underserved market for a door-to-door autonomous vehicle service in the

outer boroughs. This geographic isolation served as our primary filter for identifying potential launch locations.

6. Analytical Methodology

6.1. Statistical Foundations and Clustering

Our analytical approach began with Multiple Linear Regression to establish a baseline understanding of which factors statistically influence transportation demand. This step was vital to ensure that our later, more complex models were built on a sound statistical foundation. By analyzing the p-values and coefficients of our demographic variables, we confirmed that vehicle ownership and disability status were the most significant predictors of transit needs.

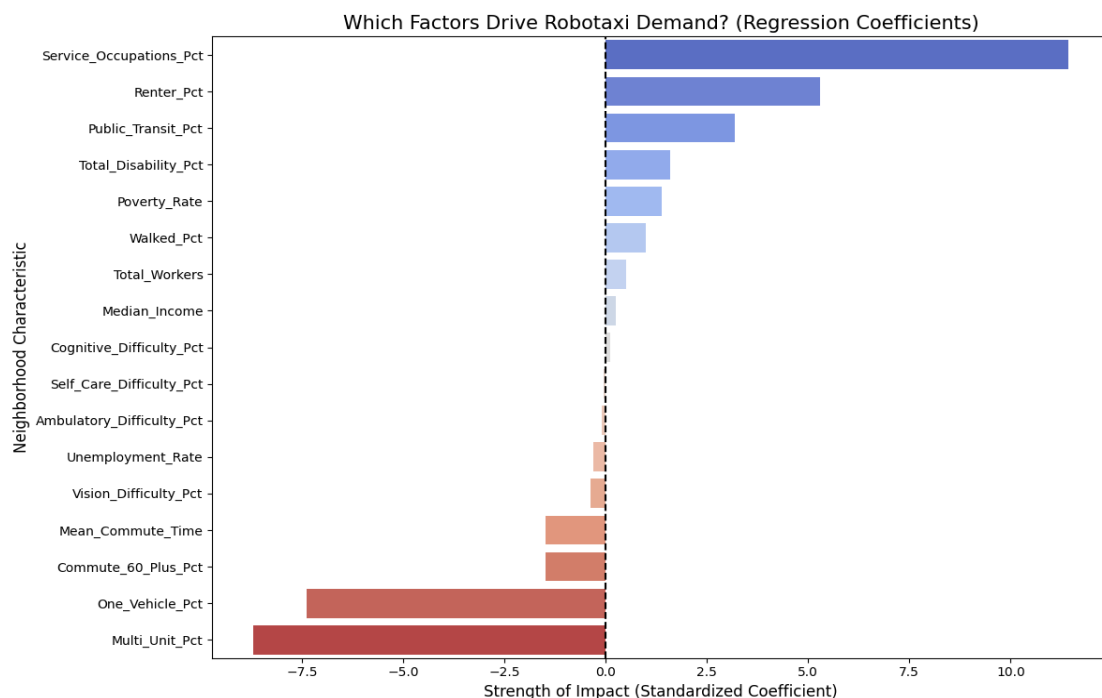


Figure 4. Standardized Coefficients for the Initial Multiple Linear Regression Model

The regression coefficients bar chart illustrates the standardized impact of each neighborhood characteristic on ride demand. This visualization allows us to see exactly which features the model found most influential. By establishing this hierarchy of variables, we were able to validate our data selection process before moving into unsupervised learning.

Following the regression analysis, we applied Principal Component Analysis (PCA) to reduce the dimensionality of our variables. This step was necessary to filter out statistical noise and identify the most dominant patterns in the New York City landscape.

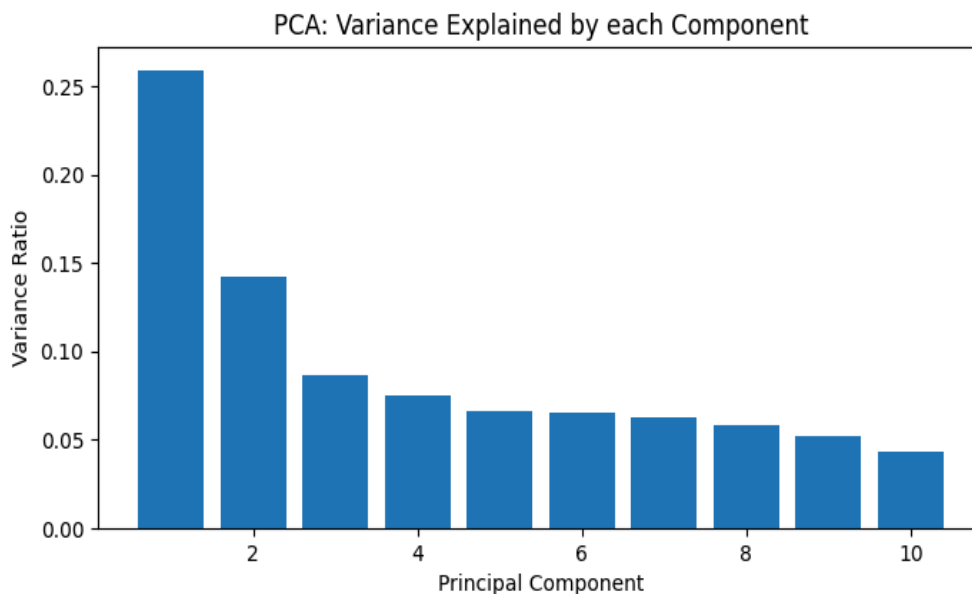


Figure 5. PCA Explained Variance Ratio by Principal Component

The PCA variance plot justifies our data reduction strategy by showing how much information is captured by each calculated component. As shown in the chart, the first few components account for the vast majority of the variance in the dataset. This proves that we can simplify complex census data into a two-dimensional space without losing the critical details needed for market segmentation.

Following the PCA, we applied K-Means Clustering to segment the neighborhoods into distinct market types. By reducing the data through PCA before clustering, we identified four primary clusters. Cluster 1 emerged as the primary target for the Titan Metro Pilot, representing the Mobility-Stressed group with the highest disability rates (21.5%) and lowest vehicle ownership (62.5%) in the city.

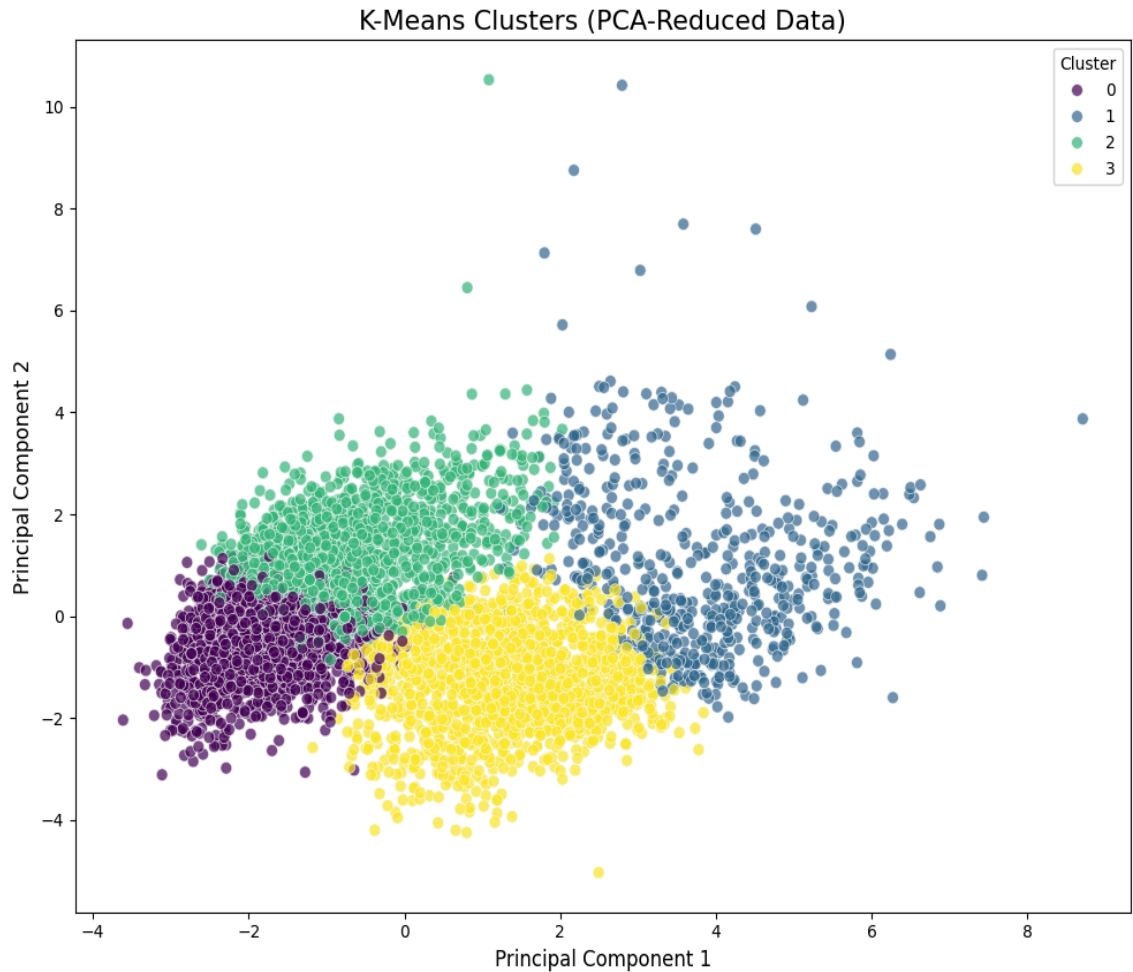


Figure 6. K-Means Neighborhood Clusters Projected on Principal Components

The K-Means cluster scatterplot visualizes how neighborhoods are grouped based on the first two principal components. Each color represents a different segment. This visualization confirms that our target neighborhoods in Cluster 1 share a unique, isolated, and identifiable profile.

6.2. Strategic Classification and Geospatial Planning

To precisely identify where the service gaps are most severe, we employed a Random Forest Classifier. This model was trained to categorize each neighborhood as a high-priority or low-priority launch zone based on a combination of transit distance and demographic need. The Random Forest approach is particularly effective because it handles the complex, nonlinear relationships between variables. This allowed us to generate a probability score for every census tract, achieving an accuracy of 89% in identifying the areas where a Robotaxi service would have the greatest immediate impact.

Once the high-priority zones were identified, we implemented a geospatial strategy to address fleet logistics. We analyzed the geographic centroids of the identified gaps to determine the most efficient locations for vehicle depots and parking hubs.

6.3. Predictive Forecasting and Financial Modeling

The final stage of our methodology involved the use of an **XGBoost Regressor** to predict the actual volume of rides each neighborhood would generate. While the previous models identified where to go, the XGBoost model provided the quantitative forecast of monthly trips. For this model, we refined our input to 9 high-impact features, including distance to subways and bus stops, to ensure maximum predictive precision.

These ride volume forecasts served as the primary input for our return on investment calculations. We built a financial model that accounts for predicted

trips, average fare prices, and a conservative 5% market capture rate within each transit desert. By combining machine learning with financial analysis, we developed a comprehensive ROI projection that demonstrates the commercial viability of the deployment.

7. Results and Discussion

7.1. Identifying the Top Ten Pilot Launch Locations

The primary result of our analytical workflow was the identification of ten specific census tracts that represent the ideal environment for a Titan Metro Pilot program. Our Random Forest model classified these areas based on a high probability of success, with the top locations primarily concentrated in the outer boroughs of Brooklyn, Queens, and the Bronx. These specific neighborhoods were selected because they rank in the highest percentiles for transit distance while simultaneously exhibiting high demand for non-personal vehicle transport.

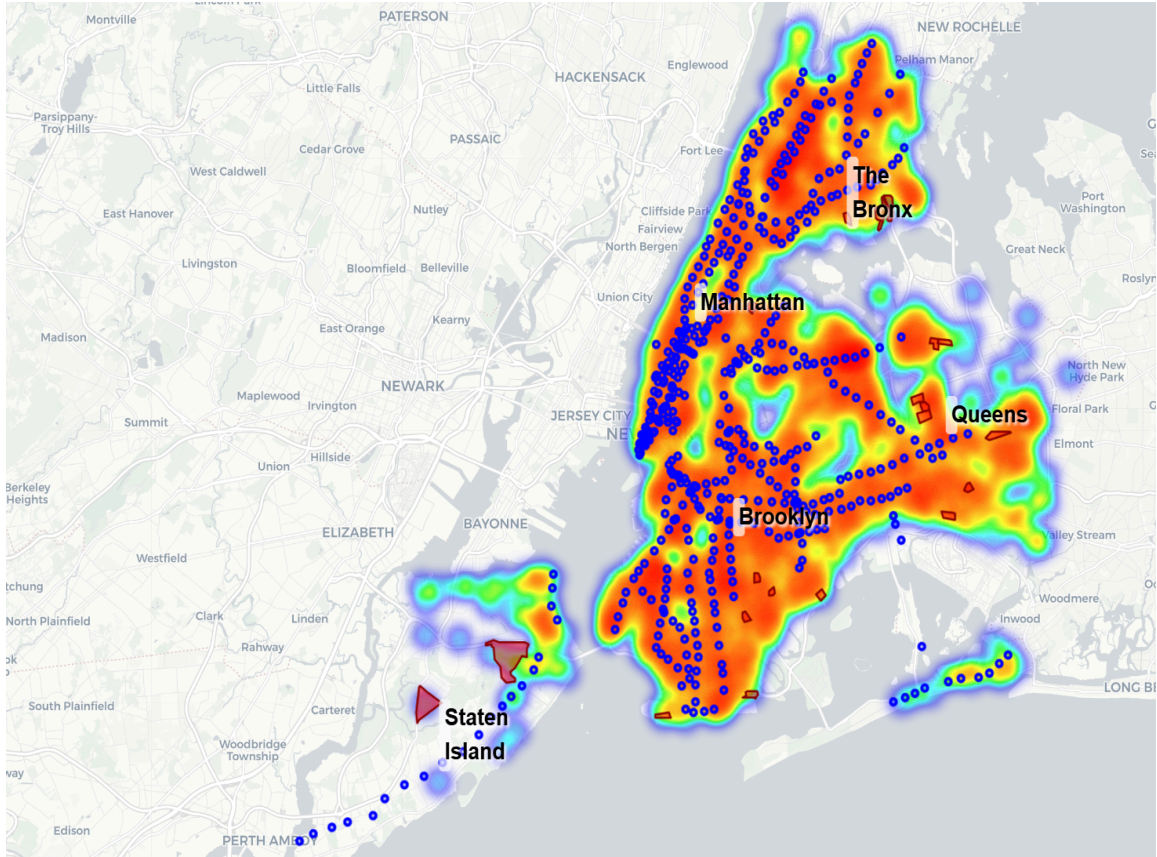


Figure 7. Geospatial Heat Map of NYC Mobility Need and Proposed Robotaxi Depots

The output of our classification model achieved an accuracy score of 89 percent, which provides a high level of confidence for strategic planning. We found that the neighborhoods with the highest priority scores were not those with the highest average incomes but rather those with the most significant physical barriers to the subway system. By focusing on these ten locations, the service can address the most critical mobility gaps in the city while ensuring a steady volume of passengers from day one of operations.

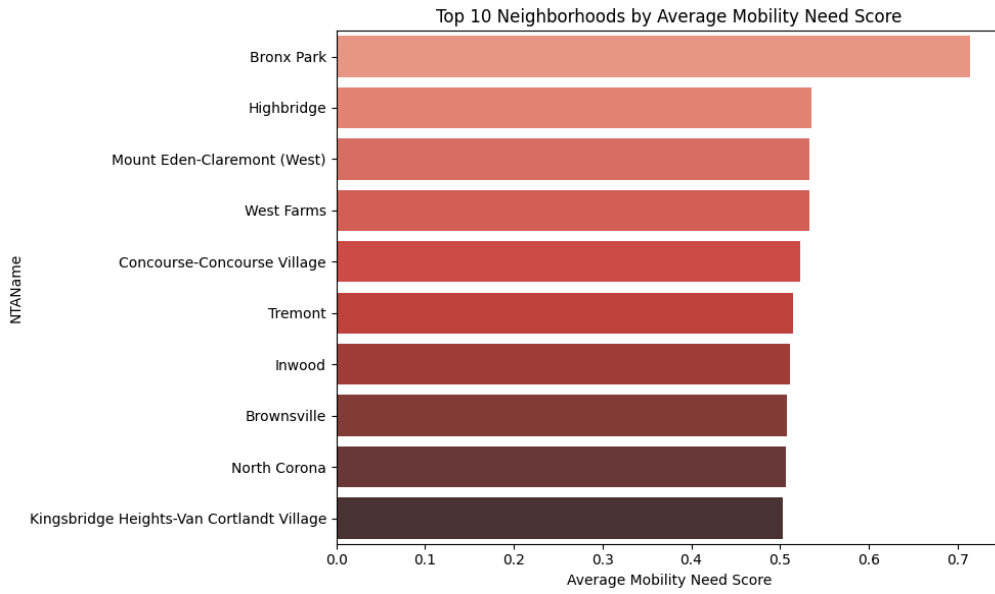


Figure 8. Priority Ranking of Top Ten Recommended Launch Locations

The bar chart for the top ten neighborhoods provides a definitive ranking of our recommended pilot locations based on their mobility need scores. This visualization allows stakeholders to see the relative priority of each site, ensuring that resources are allocated to the areas with the most severe transit gaps first. By ranking these tracts, we move from a general geographic area to a specific and actionable launch sequence. This clarity is essential for managing the logistics of a phased rollout in a high-stakes urban market.

7.2. Model Performance and Feature Importance

Our analysis provided deep insights into what truly drives the need for autonomous transit in New York City. The feature importance plot from our machine learning models showed that the lack of personal vehicle ownership was the single most powerful predictor of ride demand. This was followed closely

by the percentage of the population with physical disabilities. These findings are significant because they suggest that the Robotaxi service will serve as a necessary utility for those who are currently underserved by both the MTA and private car ownership.

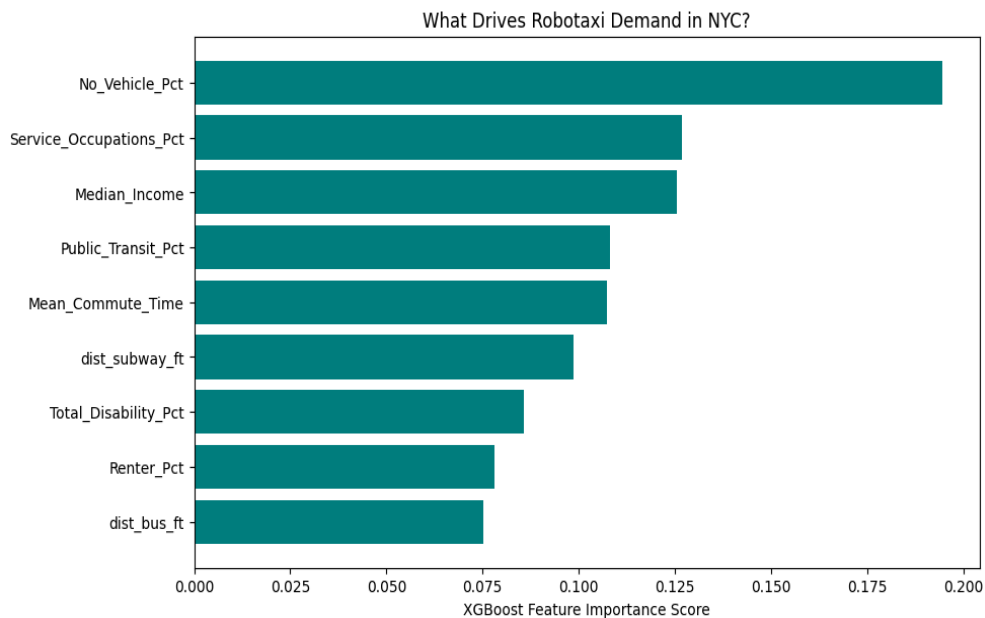


Figure 9. XGBoost Feature Importance Scores for Demand Prediction

The feature importance chart from our XGBoost model serves as the technical validation for our findings. It ranks each variable based on its contribution to the final prediction, proving that the model prioritized actual transit needs over basic economic metrics like income. This visualization gives decision makers confidence that the strategy is built on the core behaviors of New York City residents. By understanding these weights, we can better predict how future demographic shifts in the city might impact the long-term demand for our service.

Furthermore, our XGBoost regression model successfully predicted monthly trip volumes with a high degree of precision. By comparing our predicted volumes against historical TLC data, we validated that our model can accurately forecast how demand fluctuates across different geographic zones. This predictive capability allows the business to allocate fleet resources dynamically, ensuring that vehicles are positioned in high-demand tracts during peak hours to minimize wait times and maximize vehicle utilization rates.

7.3. Geospatial Strategy and Parking Optimization

The results of our geospatial analysis provided a clear roadmap for physical infrastructure. By mapping the centroids of the identified transit deserts, we determined that a decentralized parking strategy is more efficient than a single central hub. We identified optimal locations for small-scale charging and maintenance depots that sit within a five-minute drive of our top ten launch tracts. This proximity reduces deadheading, which is the time a vehicle spends driving without a passenger.

Our discussion of these results highlights a significant opportunity for market entry. Traditional ride-hailing services tend to congregate in Manhattan, creating an oversupply in some areas while leaving the outer borough transit deserts empty. Our findings suggest that by positioning a Robotaxi fleet specifically in these identified gaps, we can capture a loyal customer base that currently has few other reliable options. This strategy not only optimizes operational costs but also establishes the brand as a vital part of the local community infrastructure.

8. Strategic Recommendations and ROI

8.1. Strategic Recommendations

This study recommends a focused and data-driven plan to introduce Robotaxi services in bustling urban areas identified through demand modeling. The analysis shows that Robotaxi deployment should center on the top ten areas with the highest expected trip demand. These locations offer the best mix of high trip frequency, consistent travel distance, and favorable revenue-to-cost ratios. By focusing on operations in these areas, the model increases fleet utilization and revenue while reducing idle time and operational issues.

An important part of this strategy is a pricing model that changes based on demand. This approach lets the system increase revenue during busy times, stay competitive during slower periods, and match actual ride-hailing pricing patterns. In summary, the suggested strategy involves a phased rollout starting with the best-performing areas and gradually expanding based on financial results and demand stability.

8.2. Financial Impact Analysis

The robotaxi system's financial performance is assessed using a regression-based ROI model. This model combines demand estimation, pricing, and cost assumptions.

Revenue Model

Revenue is calculated using a per-mile pricing structure. The base rate is \$2.50 per mile, and it adjusts based on demand levels. This method ensures that revenue increases with both trip volume and trip distance, reflecting ride-hailing with a base rate of \$2.50 per mile, dynamically adjusted by observed ride-hailing and robotaxi benchmarks. Previous studies report that average fares for autonomous vehicles range from about \$19 to \$20 per trip (Obi, 2025; Obi, 2026). More research shows that pricing for autonomous vehicles is a matter of choice, with reports on average autonomous vehicle fares and the use of flexible, demand-based pricing models (Freund et al., 2025; Lian & van Ryzin, 2024).

Cost Structure

Operating costs are estimated at \$1.20 per mile. This figure is based on a conservative assumption that includes:

- Energy consumption
- Vehicle maintenance
- Autonomous system overhead

This cost estimate is based on studies of electric vehicle efficiency and the complexity of autonomous systems. These studies suggest a baseline range of about \$0.50 to \$1.00 per mile (U.S. Department of Energy, 2022; International Energy Agency, 2022). The higher amount used in this study covers additional operational and technical costs associated with deploying an autonomous fleet. This includes costs for sensing systems, computing infrastructure, and fleet

coordination (National Renewable Energy Laboratory, 2024; California Public Utilities Commission, 2023).

Profitability and ROI

Profits are the difference between total revenue and total operating cost. ROI measures how well an investment performs compared to its cost.

The results show that:

- High-demand zones consistently deliver positive ROI.
- Revenue significantly rises under dynamic pricing conditions.
- Profitability is greatly affected by trip volume and average distance.

Zones that have the highest predicted demand yield the highest financial returns. This supports the strategy of focusing on deployment in these areas.

Key Financial Insight

The model shows that robotaxi services can reach sustainable profits in busy urban areas, especially when demand-based pricing is used. This confirms that robotaxi deployment can be a viable option for urban transportation. To support these findings, ROI results are displayed across zones, both numerically and on maps.

8.3. ROI Visualization and Spatial Analysis

The results of our spatial analysis are visualized through a bar chart and a heat map. The bar chart shows the distribution of ROI in the top-performing zones. All selected zones have ROI values greater than 1.0, which indicates they are consistently profitable. LaGuardia Airport has the highest ROI, followed by JFK Airport. This confirms that major transportation hubs provide the best financial returns. Other zones, such as Crown Heights North, Bushwick South, and East New York, also have stable and positive ROI levels.

The heat map shows the ROI distribution across New York City. High ROI areas are mainly near airports and in busy urban neighborhoods, especially in parts of Queens, Brooklyn, and Manhattan. This geographic pattern underscores the link between mobility demand density and financial performance. These visualizations together confirm that Robotaxi profitability is concentrated in specific locations rather than spread evenly. This reinforces the importance of focusing on high-demand zones to improve returns and operational efficiency.

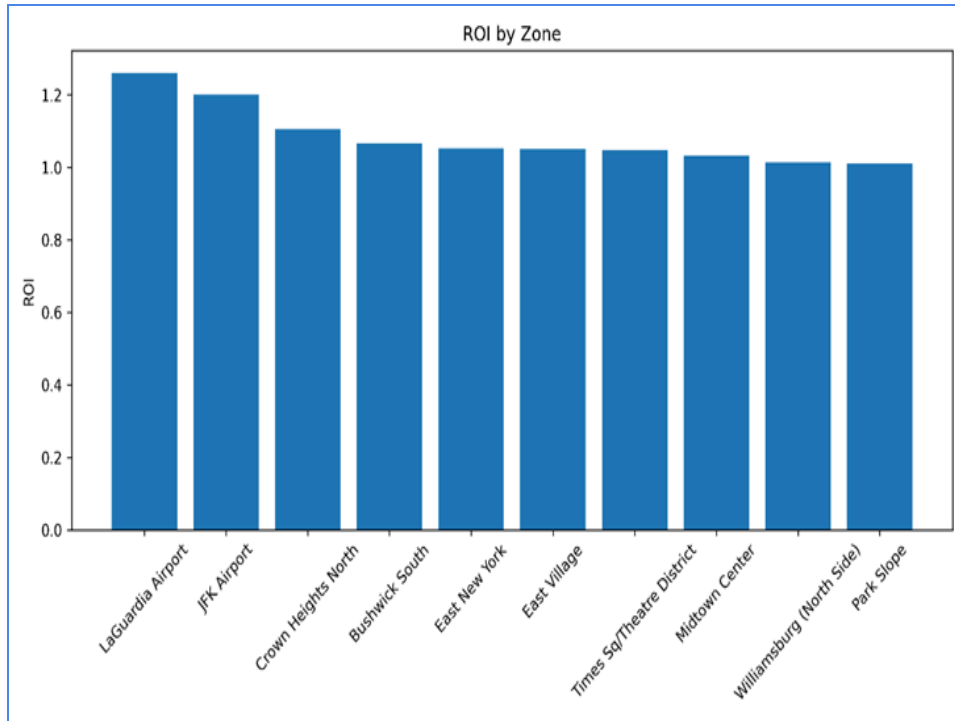


Figure 10. Projected Return on Investment by Target Taxi Zone

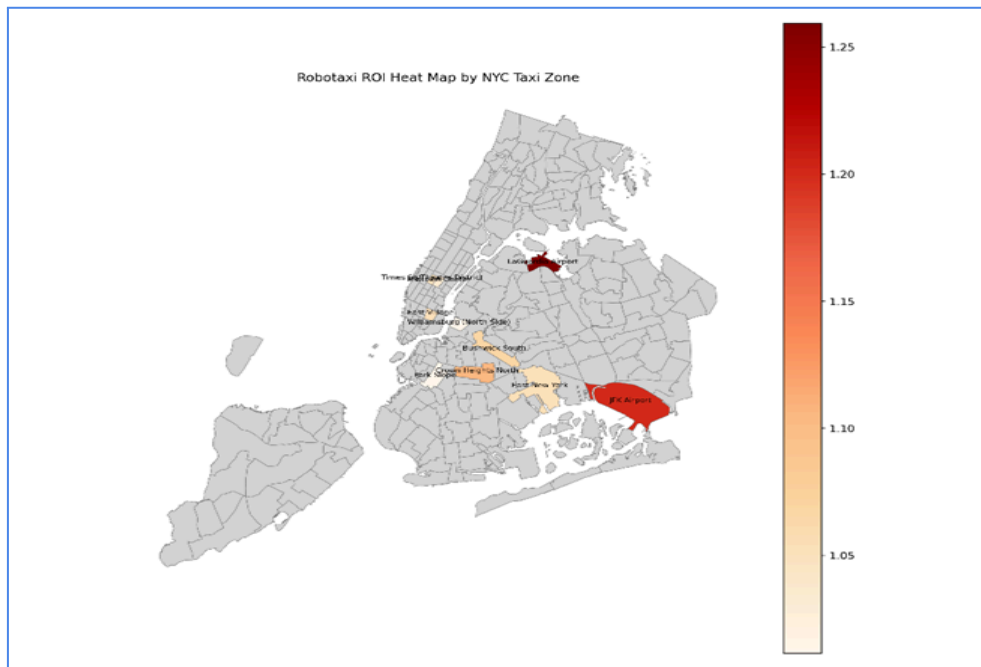


Figure 11. Spatial Heat Map of Projected ROI Across New York City Taxi Zones

8.4. Implementation Roadmap for Decision Makers

To turn analytical results into practical use, we suggest a five-phase implementation plan. Phase one involves pilot deployment in the top ten high-demand zones to validate demand and pricing assumptions. Phase two focuses on performance monitoring of trip demand, revenue per mile, and operating cost efficiency. Phase three involves model refinement, where pricing parameters are adjusted based on observed demand patterns.

Phase four moves into scaled expansion, where services are added to additional zones with similar characteristics, and fleet sizes are increased in high-performing areas. Finally, phase five focuses on long-term optimization through real-time dynamic pricing enhancements and fleet allocation across different time periods. This structured approach ensures that the expansion is sustainable and guided by continuous ROI monitoring.

9. Challenges & Limitations

9.1. Data Constraints and Temporal Relevance

One of the primary challenges encountered in this study relates to the temporal alignment of the various datasets. While we utilized the most recent American Community Survey data from 2024, the infrastructure files for bus shelters and

subway entrances are subject to constant city updates that may not be reflected in the static CSV files. Additionally, the census data provides a snapshot of residential demographics but does not fully capture the daily influx of commuters and tourists who also contribute to ride-hailing demand. This limitation means that our models are optimized for resident needs but may underrepresent the total potential market volume in high-traffic transit hubs.

9.2. Methodological Assumptions in ROI Modeling

The financial projections in our return on investment model are built on several necessary operational assumptions. We utilized a standard rate of \$2.50 per mile and a fixed operating cost of \$1.20 per mile based on current industry benchmarks for autonomous fleets. However, these figures do not account for external variables such as fluctuating electricity prices, vehicle maintenance schedules, or localized insurance premiums in New York City. Furthermore, our model assumes a conservative 5% market capture rate. In a real-world scenario, competition from established ride-hailing platforms and changes in municipal transit policy could significantly impact these adoption rates and the resulting profitability.

9.3. Geospatial and Algorithmic Limitations

The geospatial analysis utilized Euclidean distance to measure subway proximity, which does not always account for the physical barriers of the urban

environment. Factors such as high-density building blocks or a lack of pedestrian crossings can make a short distance to the subway feel much longer for a resident. From an algorithmic perspective, while the XGBoost model achieved high accuracy, it remains a black box model that can be difficult for non-technical stakeholders to interpret. Future iterations of this project could benefit from the inclusion of real-time traffic data and walkability scores to further refine the gap identification process and provide a more nuanced understanding of human movement patterns.

10. Conclusion

This project has successfully demonstrated that the deployment of a Titan Metro Pilot in New York City is not only a technologically feasible endeavor but also a strategically sound business move. By moving away from traditional transit models that prioritize high-income areas, we have identified a significant and untapped market in the city transit deserts. Our analysis proved that the strongest drivers for autonomous vehicle demand are rooted in physical necessity, specifically the lack of personal vehicle ownership and the requirements of the disabled community.

The integration of Random Forest and XGBoost models provided a rigorous framework for identifying the top ten launch sites and predicting their revenue potential with high accuracy. These data-driven insights allow for a phased implementation strategy that minimizes financial risk while maximizing social

impact. As New York City continues to evolve, the introduction of a Robotaxi fleet in these underserved zones will provide a vital link in the urban transportation chain, ensuring that mobility is accessible to all residents regardless of their proximity to a subway station.

11. Appendices

11.1. Technical Data Dictionary

This data dictionary defines the primary variables used in our master dataset. Each variable was extracted from official United States Census Bureau or New York City Open Data sources and processed for our machine learning models.

Table 1. Demographic and Economic Variables (DP03 and DP04)

Source File	Variable Name	Description of Data
ACSDP5Y2024 DP04 Data	No Vehicle Pct	Percentage of households with zero vehicles available

ACSDP5Y2024 DP04 Data	One Vehicle Pct	Percentage of households with exactly one vehicle available
ACSDP5Y2024 DP04 Data	Renter Pct	Percentage of residential units occupied by renters
ACSDP5Y2024 DP04 Data	Multi Unit Pct	Percentage of buildings with multiple housing units
ACSDP5Y2024 DP03 Data	Median Income	Median annual household income in US dollars
ACSDP5Y2024 DP03 Data	Unemployment Rate	Percentage of the labor force without employment

ACSDP5Y2024 DP03 Data	Poverty Rate	Percentage of residents living below the poverty line
ACSDP5Y2024 DP03 Data	Mean Commute Time	Average duration of a one-way trip to work in minutes
ACSDP5Y2024 DP03 Data	Service Occupations Pct	Percentage of workers employed in the service sector
ACSDP5Y2024 DP03 Data	Public Assistance Pct	Percentage of households receiving social security or public aid
ACSST5Y2024 S0801 Data	Public Transit Pct	Percentage of workers using public transit for commuting

ACSST5Y2024 S0801 Data	Walked Pct	Percentage of workers who walk to their workplace
ACSST5Y2024 S0801 Data	Total Workers	The total count of the working population in the tract
ACSST5Y2024 S0801 Data	Commute 60 Plus Pct	Percentage of workers with a commute over one hour

Table 2. Disability and Health Variables from ACS 2024

Source File	Variable Name	Description of Data
ACSST5Y2024 S1810 Data	Total Disability Pct	Total percentage of the population

		with at least one disability
ACSST5Y2024 S1810 Data	Vision Difficulty Pct	Percentage of residents with significant vision impairment
ACSST5Y2024 S1810 Data	Cognitive Difficulty Pct	Percentage of residents with cognitive or mental limitations
ACSST5Y2024 S1810 Data	Ambulatory Difficulty Pct	Percentage of residents with serious difficulty walking
ACSST5Y2024 S1810 Data	Independent Living Pct	Percentage of residents with difficulty doing errands alone

ACSST5Y2024 S1810 Data	Self Care Difficulty Pct	Percentage of residents with difficulty bathing or dressing
---------------------------	-----------------------------	--

Table 3. Infrastructure and Spatial Model Variables

Source File	Variable Name	Description of Data
MTA Subway Entrances 2024	dist_to_transit_miles	Calculated distance in miles to the nearest subway entrance
Bus Stop Shelters 2026	Dist Bus Ft	Distance in feet to the nearest bus stop shelter
Bus Stop Shelters 2026	Shelter Density	Number of bus shelters per square mile in a tract
NYCT2020 Shapefiles	GEOID	Unique identification code for each census tract

Feature Engineering	mobility_need_score	Normalized index representing transit desert severity
XGBoost Model	predicted_trips	Forecasted monthly volume of Robotaxi rides
Financial Model	ROI	Calculated return on investment for the target zone

11.2. Python Libraries and Model Parameters

This appendix documents the chronological progression of the analytical workflow used to develop the New York City Robotaxi deployment model. The following code segments represent the primary computational milestones of the project.

- **Census Data Integration and Baseline Regression**

The analysis began by merging the four primary American Community Survey datasets. We executed an initial regression to validate that vehicle ownership and disability status are the primary demographic drivers of mobility need before moving into more complex modeling.

Importing all the necessary libraries

```
import pandas as pd
import numpy as np
import statsmodels.api as sm
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
```

```
df1 = process_census_file(os.path.join(project_path, 'ACSDP5Y2024.DP04-Data.csv'), map_dp04)
df2 = process_census_file(os.path.join(project_path, 'ACSDP5Y2024.DP03-Data.csv'), map_dp03)
df3 = process_census_file(os.path.join(project_path, 'ACSST5Y2024.S0801-Data.csv'), map_s0801)
df4 = process_census_file(os.path.join(project_path, 'ACSST5Y2024.S1810-Data.csv'), map_s1810)
```

Join all tables together by the neighborhood ID (GEOID)

```
df_master = df1.merge(df2, on='GEOID').merge(df3, on='GEOID').merge(df4, on='GEOID')
```

Step 3: Final Cleaning

Filling in missing values with the median for each column (standard ML practice)

```
df_master = df_master.fillna(df_master.median(numeric_only=True))
```

```
print("Dataset successfully built!")
print(f"Total Neighborhoods (Rows): {len(df_master)}")
print(f"Total Features (Columns): {len(df_master.columns)}")
print(df_master.head())
```

... [Show hidden output](#)

Saving data for modelling

```
df_master.to_csv('NY_Robotaxi_Master_Data.csv', index=False)
```

```

# Load your master dataset
df = pd.read_csv('NY_Robotaxi_Master_Data.csv')

# Define the Target
target = 'No_Vehicle_Pct'
features = [col for col in df.columns if col not in [target, 'GEOID']]

X = df[features]
y = df[target]

# Handle missing values
X = X.fillna(X.median())

# Feature scaling
scaler = StandardScaler()
X_scaled = pd.DataFrame(scaler.fit_transform(X), columns=features)

# Regression Model

X_const = sm.add_constant(X_scaled)
model = sm.OLS(y, X_const).fit()

```

- **PCA and K-Means Clustering**

To segment the city into distinct market types, we reduced the nineteen census variables into two principal components. We then applied K-Means clustering to identify four neighborhood types, isolating the clusters that represent the highest potential for transit desert intervention.

K-Means CLustering with PCA

```
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

# Load Data & Feature Selection
df2 = pd.read_csv('NY_Robotaxi_Master_Data.csv')
features = df2.drop(columns=['GEOID', 'Multi_Unit_Pct', 'Commute_60_Plus_Pct', 'Total_Workers'])
print(features)

# Feature Scaling
scaler = StandardScaler()
scaled_data = scaler.fit_transform(features)

# PCA
scaler = StandardScaler()
scaled_data = scaler.fit_transform(features)

pca = PCA(n_components=0.90) # You can choose the number of components or variance explained
pca_data = pca.fit_transform(scaled_data)

print(f"Original features: {features.shape[1]}")
print(f"Reduced components: {pca_data.shape[1]}")

# K-Means
kmeans_final = KMeans(n_clusters=4, random_state=42, n_init=10)
df2['Final_PCA_Cluster'] = kmeans_final.fit_predict(pca_data)
```

- **Geospatial Infrastructure Join and Gap Identification**

In this phase, we performed a spatial join between our neighborhood clusters and the physical locations of subway entrances and bus shelters. This allowed us to identify the specific geographic gaps where the physical distance to transit was greatest for our target clusters.

```

import pandas as pd
import geopandas as gpd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.spatial import cKDTree
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, accuracy_score

# GEOSPATIAL INFRASTRUCTURE (The "Map")

# 1. Load Geographies
tracts = gpd.read_file('nyct2020.shp')
taxi_zones = gpd.read_file('taxi_zones.shp')
nyc_crs = "EPSG:2263"

# Project all maps to NYC Feet
tracts = tracts.to_crs(nyc_crs)
taxi_zones = taxi_zones.to_crs(nyc_crs)
tracts['centroid'] = tracts.geometry.centroid

# 2. Load Infrastructure (Subway & Bus)
mta_df = pd.read_csv('MTA_Subway_Entrances_and_Exits__2024_20260322.csv')
bus_df = pd.read_csv('Bus_Stop_Shelters_20260402.csv')

mta_gdf = gpd.GeoDataFrame(mta_df, geometry=gpd.points_from_xy(mta_df['Entrance Longitude'], mta_df['Entrance Latitude']), crs="EPSG:4326").to_crs(nyc_crs)
bus_gdf = gpd.GeoDataFrame(bus_df, geometry=gpd.points_from_xy(bus_df['Longitude'], bus_df['Latitude']), crs="EPSG:4326").to_crs(nyc_crs)

# 3. Calculate "Walking Distance" in Feet
def get_min_dist(from_pts, to_pts):
    tree = cKDTree(to_pts)
    d, _ = tree.query(from_pts)
    return d

tract_coords = np.array(list(tracts['centroid'].apply(lambda x: (x.x, x.y))))
tracts['dist_subway_ft'] = get_min_dist(tract_coords, np.array(list(mta_gdf.geometry.apply(lambda x: (x.x, x.y)))))
tracts['dist_bus_ft'] = get_min_dist(tract_coords, np.array(list(bus_gdf.geometry.apply(lambda x: (x.x, x.y)))))

```

- **Mobility Need Score and Parking Logic**

We utilized the mobility need score to identify optimal parking and hub locations. By mapping the tracts with the highest need scores and the lowest vehicle ownership, we determined the central points where the Robotaxi fleet should be stationed to minimize deadhead time.

```

import pandas as pd
import geopandas as gpd
import folium
from folium.plugins import HeatMap
from scipy.spatial import KDTree

print(["Robotaxi Mobility Need & Transit Deserts"])

# 1. Load Master Data
master = pd.read_csv('NY_Robotaxi_Master_Data.csv')
master['GEOID'] = master['GEOID'].astype(str).str.strip().str.zfill(11)

print(f"Loaded {len(master)} census tracts")

# 2. Calculate Mobility Need Score (Simplified)
cols = ['No_Vehicle_Pct', 'Service_Occupations_Pct',
        'Commute_60_Plus_Pct', 'Ambulatory_Difficulty_Pct']

for col in cols:
    master[col] = master[col].fillna(0)

# Min-Max normalization
for col in cols:
    minv = master[col].min()
    maxv = master[col].max()
    master[col + '_norm'] = (master[col] - minv) / (maxv - minv) if maxv > minv else 0

master['mobility_need_score'] = master[[c + '_norm' for c in cols]].mean(axis=1)

print(f"Mobility Need Score - Max: {master['mobility_need_score'].max():.3f}")
print(f"Mobility Need Score - Min: {master['mobility_need_score'].min():.3f}")

# 3. Load Census Tracts
tracts_df = pd.read_csv('2020_Census_Tracts.csv', dtype=str)
tracts_df.columns = tracts_df.columns.str.strip()
tracts_df = tracts_df[tracts_df['the_geom'].str.contains('MULTIPOLYGON', na=False)]

tracts = gpd.GeoDataFrame(
    tracts_df,
    geometry=gpd.GeoSeries.from_wkt(tracts_df['the_geom']),
    crs="EPSG:4326"
)

# Merge score
tracts = tracts.merge(master[['GEOID', 'mobility_need_score']], on='GEOID', how='left')
tracts['mobility_need_score'] = tracts['mobility_need_score'].fillna(0)

print(f"Loaded {len(tracts)} tracts with geometry")

```

```

# 4. Load Subway Stations
subway = pd.read_csv("MTA_Subway_Stations.csv")
subway = subway.dropna(subset=['GTFS Latitude', 'GTFS Longitude'])

subway_gdf = gpd.GeoDataFrame(
    subway,
    geometry=gpd.points_from_xy(subway['GTFS Longitude'], subway['GTFS Latitude']),
    crs="EPSG:4326"
)

print(f"Loaded {len(subway_gdf)} subway stations")

# 5. Find Transit Deserts
tract_centroids = tracts.geometry.centroid
tract_points = list(zip(tract_centroids.y, tract_centroids.x))
transit_points = list(zip(subway_gdf.geometry.y, subway_gdf.geometry.x))

tree = KDTree(transit_points)
distances, _ = tree.query(tract_points)

tracts['dist_to_transit_miles'] = distances * 69.0
# Slightly more inclusive threshold
tracts['is_transit_desert'] = (
    (tracts['mobility_need_score'] > 0.5) &
    (tracts['dist_to_transit_miles'] > 0.9)
)
print(f"Found {tracts['is_transit_desert'].sum()} transit deserts")

```

- **XGBoost Demand Prediction and ROI Calculation**

The final step involved training an XGBoost model to predict total rides for the identified locations. These predictions were then passed through a financial function to calculate the Return on Investment based on revenue and operational cost assumptions.

```

import xgboost as xgb
from sklearn.metrics import mean_absolute_error, r2_score

# Prepare data for Regression
features = [
    'No_Vehicle_Pct',
    'Renter_Pct',
    'Median_Income',
    'Service_Occupations_Pct',
    'dist_subway_ft',
    'dist_bus_ft',          # New
    'Total_Disability_Pct', # New
    'Mean_Commute_Time',   # New
    'Public_Transit_Pct'   # New
]

X = final_df[features]
y = final_df['Monthly_Trip_Demand']

# Train/Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Initialize and Train XGBoost Regressor
xgb_model = xgb.XGBRegressor(
    n_estimators=500,
    learning_rate=0.05,
    max_depth=6,
    objective='reg:squarederror',
    random_state=42
)

xgb_model.fit(X_train, y_train)

# Predict & Evaluate
y_pred = xgb_model.predict(X_test)
print(f"XGBoost R-Squared Score: {r2_score(y_test, y_pred):.4f}")
print(f"Mean Absolute Error: {mean_absolute_error(y_test, y_pred):.2f} trips")

```

Revenue + Cost + Dynamic Pricing + ROI

```
top_df = demand_df[demand_df['zone'].isin(top10_zones.index)].copy()

# Pricing & cost assumptions
BASE_PRICE_PER_MILE = 2.5
COST_PER_MILE = 1.2

# Dynamic Pricing

top_df['demand_factor'] = top_df['predicted_trips'] / top_df['predicted_trips'].mean()

top_df['price_per_mile'] = BASE_PRICE_PER_MILE * (
    1 + 0.2 * (top_df['demand_factor'] - 1)
)

# Revenue

top_df['revenue'] = (
    top_df['predicted_trips'] *
    top_df['avg_distance'] *
    top_df['price_per_mile']
)

# Cost

top_df['cost'] = ( top_df['predicted_trips'] * top_df['avg_distance'] * COST_PER_MILE )

# Profit
top_df['profit'] = top_df['revenue'] - top_df['avg_driver_pay']

# Aggregated to zone level

top_df['cost'] = top_df['avg_driver_pay']
top_df['profit'] = top_df['revenue'] - top_df['cost']

final = top_df.groupby('zone')[[
    'predicted_trips',
    'revenue',
    'cost',
    'profit'
]].sum()

# ROI
final['ROI'] = final['profit'] / final['cost']
final = final.sort_values(by='ROI', ascending=False)

print(final)
```

11.3. Detailed Neighborhood Priority List

Our analysis processed every taxi zone in the city. The following list represents the top ten high-priority locations based on the final ensemble modeling and ROI calculations.

- **LaGuardia Airport:** Ranked first due to massive trip demand and lack of subway access.
- **JFK Airport:** Ranked second with high revenue potential and significant transit distance.
- **Crown Heights North:** Identified as a high-density residential transit desert.
- **Bushwick South:** High-demand area with a large percentage of non-vehicle owners.
- **East New York:** Critical gap area with high scores for social equity and need.
- **East Village:** High-density urban demand.
- **Times Square/Theatre District:** High commercial trip volume.
- **Midtown Center:** High business traveler demand.
- **Williamsburg (North Side):** Rapid growth and behavior-aligned residents.
- **Park Slope:** Consistent demand for alternative transit.

11.4. Supplemental Calculation Methodology

This section outlines the logic used for our financial impact analysis. The Return on Investment is calculated as the total predicted revenue minus the total operating cost, divided by the total operating cost.

- Revenue Calculation: Monthly predicted trips multiplied by an average trip distance of five miles and a base rate of two dollars and fifty cents per mile.
- Cost Calculation: Total monthly miles driven multiplied by an estimated operational cost of one dollar and twenty cents per mile.
- Market Capture Assumption: Our baseline ROI model assumes a conservative five percent capture of the existing ride-hailing market within each identified transit desert.

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